

18/11/25

Example :-

x	y	$\hat{y}$
1	6	5
3	23	19
9	60	61
10	67	68

$7x-2$

assumed ϵ_1^n given by SLR model

$$\text{Error} = AV - PV$$

Evaluation metrics of Regression :-

1. R^2 score (or) R square value (R^2 value) $\rightarrow$

How much good model it is?

(higher the better)

2. MSE (Mean Squared Error)

3. MAE (Mean Absolute Error)

4. RMSE (Root Mean Squared Error)

5. MAPE (Mean Absolute Percentage Error)

$\rightarrow$ how bad model it is?
(lower the better)

R^2 score / R^2 value :-

$$R^2 = 1 - \frac{LR}{AVG_1}$$

where

LR $\rightarrow$ Total error of predict model

AVG $\rightarrow$ Total error of ^{average} actual model

Error $\rightarrow$ Actual value - Predicted value

* Total Error of predicted model :-

$$PV_{LR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

y_i :- $\rightarrow$ actual value

$\hat{y}_i$:- $\rightarrow$ predicted value

* Total Error of Average model :-

$$AVG = \sum_{i=1}^n (\bar{y}_i - y_i)^2$$

$\bar{y}_i$:- mean value of y .

$\rightarrow$ TEOP (LR) :-

$$LR = (6-5)^2 + (23-19)^2 + (60-61)^2 + (67-68)^2$$
$$= 1 + 16 + 1 + 1$$

$$\boxed{LR = 19}$$

$\rightarrow$ TEOA (AVG) :-

$$AVG = (39-6)^2$$

$$\bar{y}_i = \frac{6+23+60+67}{4}$$

$$= \frac{156}{4} = 39$$

$$AVG = (39-6)^2 + (39-23)^2 + (39-60)^2 + (39-67)^2$$

$$\boxed{\bar{y}_i = 39}$$

$$= (33)^2 + (16)^2 + (-21)^2 + (-28)^2$$

$$= 1089 + 256 + 441 + 784$$

$$\boxed{AVG = 2570}$$

$$R^2 = 1 - \frac{LR}{AVG}$$

$$= 1 - \frac{19}{2570}$$

$$= 1 - 0.0073$$

$$R^2 = 0.9926$$

→ R^2 range is (0-1)

if R^2 value is near to 1 means, it is best model.
if R^2 value is near to 0 means, it is worst model.

Here, R^2 value is near to 1, so, the model is considered as best model.

Steps we follow in machine learning:-

1. Data Analysis (Pandas, numpy, matplotlib)

2. Data Validation

3. Model building } sklearn
 } (scikit)

→ Build a model → Algorithm name

→ Train a model → (x, y)

→ Test a model → x

→ Evaluate a model → result analysis

2. Mean Square Error (MSE):-

x	y	$\hat{y}$
1	6	5
3	23	19
9	60	61
10	67	66

$$\hat{y} = 7x - 2$$

→ MSE measure how far predictions are from a actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{4} \left[(6-5)^2 + (23-19)^2 + (60-61)^2 + (67-66)^2 \right]$$

$$= \frac{1}{4} (19)$$

$$MSE = 4.75$$

*
→ MSE ranges from 0 to ∞ , if the lower the value, then the model is better.

Here, MSE is less hence the model is good & have good accuracy.

3. Mean Absolute Error (MAE):-

MAE tells on avg how many units the model is differing.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$= \frac{1}{4} [|66-5| + |23-19| + |60-61| + |67-66|]$$

$$= \frac{1}{4} [1+4+1+1]$$

$$= \frac{7}{4}$$

$$\boxed{MAE = 1.75}$$

* MAE ranges from 0 to ∞ .

MAE value near to ∞ means greater error & vice versa.

Here, MAE value is less hence model is good.

4. Root Mean Square Error:-

It indicates error in actual units of y .

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$RMSE = \sqrt{4.75}$$

$$\boxed{RMSE = 2.179}$$

* larger RMSE means worst accuracy.

Here, RMSE is smaller hence model is having good accuracy.

5. Mean Absolute Percentage Error (MAPE):

$$\begin{aligned} \boxed{MAPE} &= \frac{100}{n} \sum_{i=1}^n \left| \frac{y - \hat{y}}{y} \right| \\ &= \frac{100}{4} \left[\left| \frac{6-5}{6} \right| + \left| \frac{23-19}{23} \right| + \left| \frac{60-61}{60} \right| + \left| \frac{67-66}{67} \right| \right] \\ &= \frac{100}{4} \left[\frac{1}{6} + \frac{4}{23} + \frac{1}{60} + \frac{1}{67} \right] \\ &= \frac{100}{4} [0.3722] \\ \boxed{MAPE} &= 9.30\% \end{aligned}$$

MAPE give percentage Error

If % Error :-

- 0% — Perfect fit
- 0-10% — Highly accurate
- 10-20% — Good
- 20-50% — Reasonable
- >50% — Poor

⇒ Here our MAPE value lies in 10-20%. Hence the model is highly accurate.